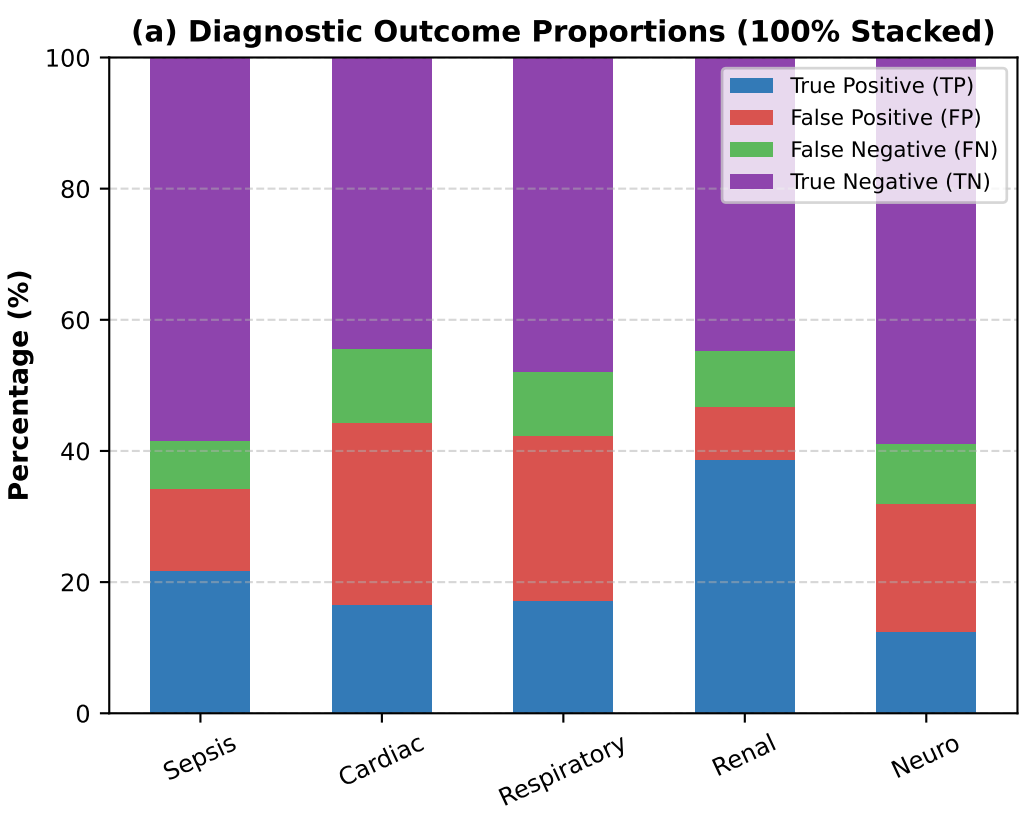
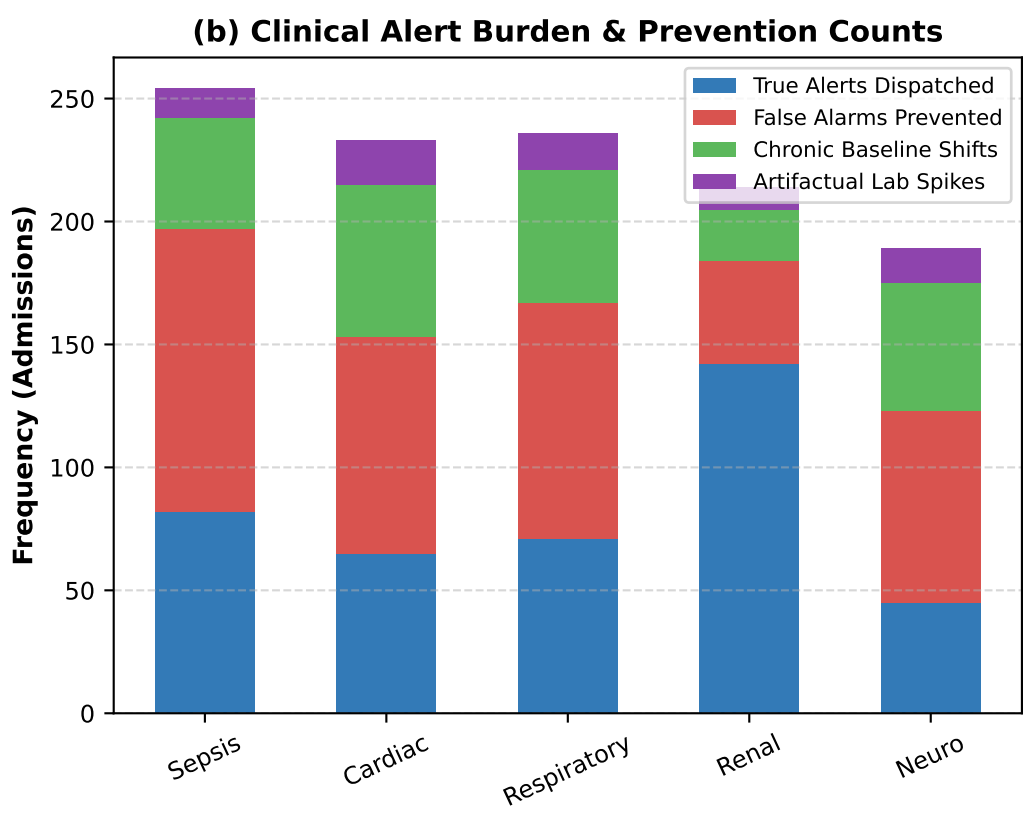
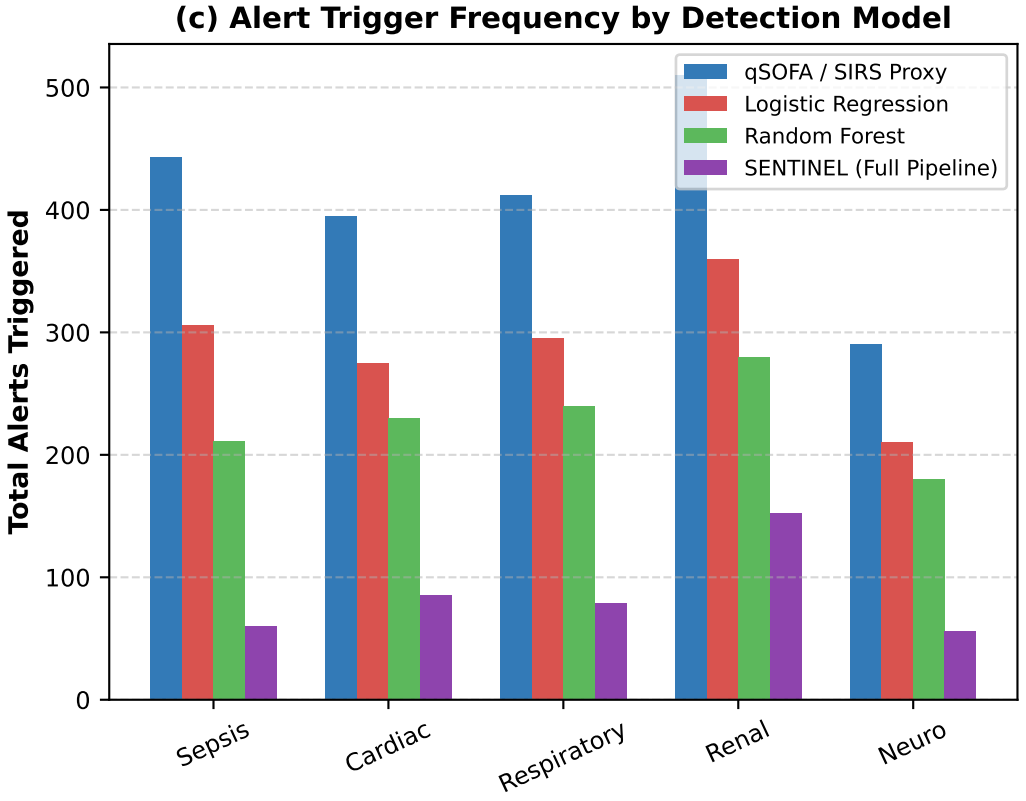
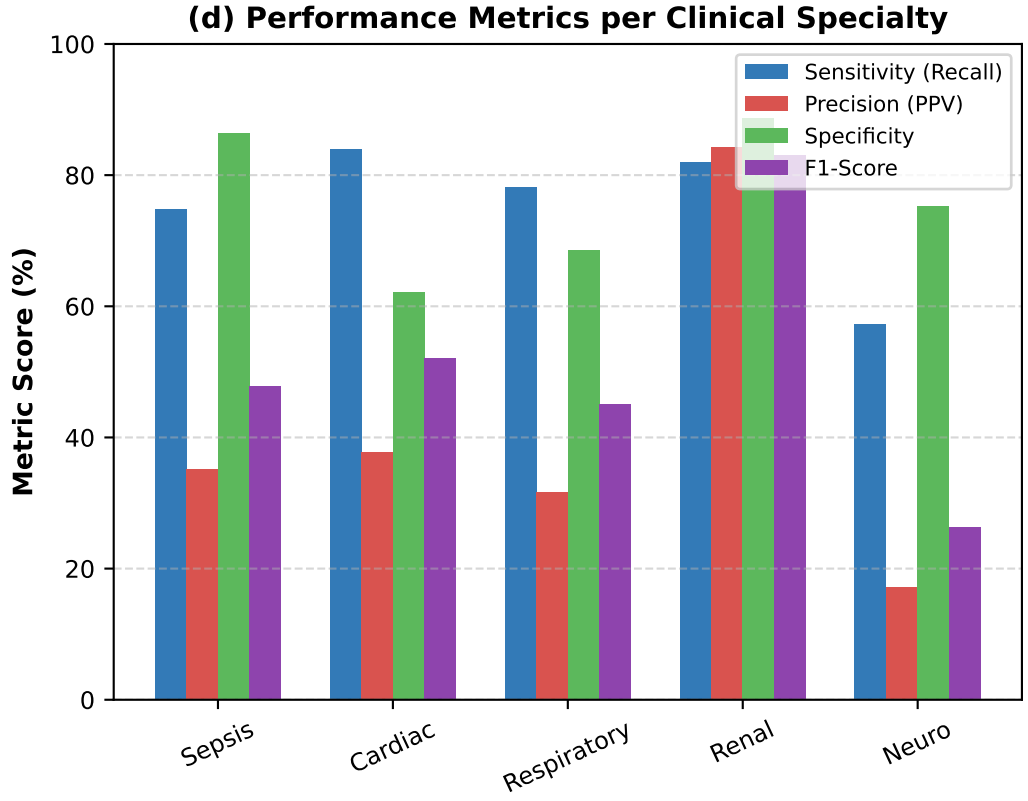
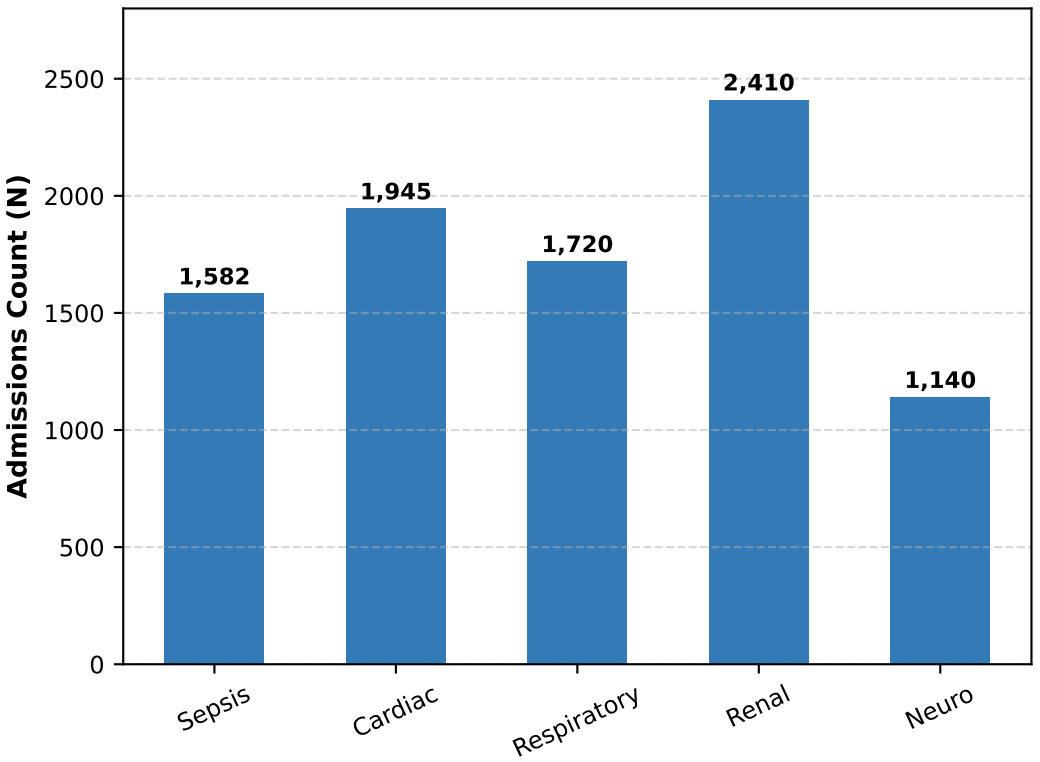
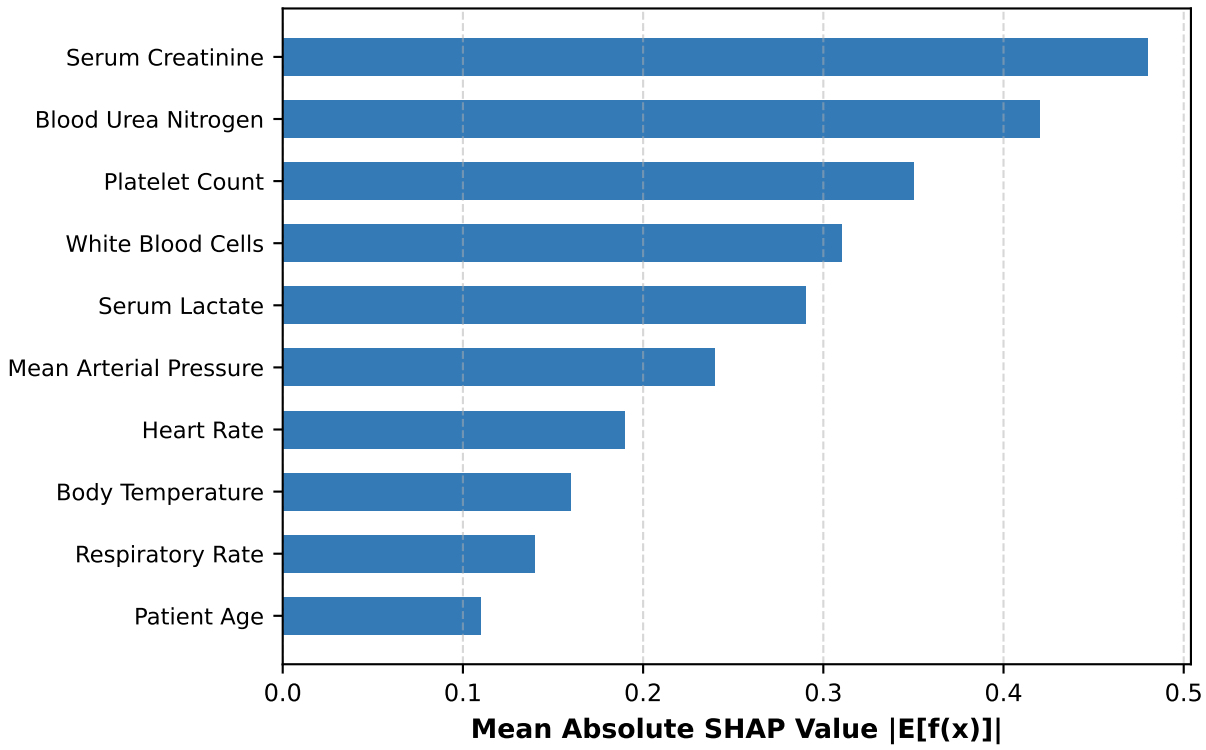

(a) Diagnostic Outcome Proportions (100% Stacked)
Percentage (%)
True Positive (TP)
False Positive (FP)
False Negative (FN)
True Negative (TN)
Sepsis
Cardiac
Respiratory
Renal
Neuro


(b) Clinical Alert Burden & Prevention Counts
Frequency (Admissions)
True Alerts Dispatched
False Alarms Prevented
Chronic Baseline Shifts
Artifactual Lab Spikes
Sepsis
Cardiac
Respiratory
Renal
Neuro


(c) Alert Trigger Frequency by Detection Model
Total Alerts Triggered
qSOFA / SIRS Proxy
Logistic Regression
Random Forest
SENTINEL (Full Pipeline)
Sepsis
Cardiac
Respiratory
Renal
Neuro


(d) Performance Metrics per Clinical Specialty
Metric Score (%)
Sensitivity (Recall)
Precision (PPV)
Specificity
F1-Score
Sepsis
Cardiac
Respiratory
Renal
Neuro


(e) Pathology Prevalence Across Working Cohort (N=7,238)
Admissions Count (N)
1,582
1,945
1,720
2,410
1,140
Sepsis
Cardiac
Respiratory
Renal
Neuro


(f) Top Physiological Biomarkers Ranked by SHAP
Serum Creatinine
Blood Urea Nitrogen
Platelet Count
White Blood Cells
Serum Lactate
Mean Arterial Pressure
Heart Rate
Body Temperature
Respiratory Rate
Patient Age
Mean Absolute SHAP Value |E[f(x)]|